

HITESH GOYAL

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ABOUT

Performance-driven AI Engineer and NTU graduate skilled in the end-to-end model lifecycle—from training and fine-tuning to managing containerized environments via Docker and AWS.

EDUCATION

Nanyang Technological University <i>Master of Science in Artificial Intelligence (CGPA: 3.89)</i>	Singapore Jan. 2026
Vellore Institute of Technology <i>B.Tech in Computer Science (Specialization in AI/ML, CGPA: 9.06)</i>	Chennai, India Jul. 2023

WORK EXPERIENCE

Graduate AI Engineering Intern (Summation Fellow) <i>Hummingbird Bioscience (SGInnovate Deep-Tech Programme)</i>	Aug. 2025 – Jan. 2026 Singapore
- Achieved a 50× runtime reduction in cancer-target discovery by implementing PyTorch-GPU vectorization and re-engineering data augmentation logic. Tripled feature deployment throughput on HPC clusters by architecting a config-driven SLURM orchestration tool for interactive GPU sessions. Orchestrated Generative AI deployment (BoltzGen) on AWS EC2 using DL AMIs for stable protein binder inference. 90% reduction in deck-creation time (2 hrs to 10 mins) through a custom interactive visualization system with persistent dashboards.	
AI Engineer Full Time (Computer Vision) <i>Tata Elxsi</i>	Jul. 2023 – Jun. 2024 Bangalore, India
Secured a new client by engineering a real-time CV/clustering anomaly detection pipeline for pollution monitoring. - Achieved 90% recall and 80% precision in content moderation using a DP-inspired temporal smoothing algorithm; parallelized it with Celery for distributed queuing. - Enabled lag-free ad placement at 30+ FPS with stream-agnostic inference logic ; boosted object tracking throughput by 20× via batching. - Achieved 78.1% mAP on industrial defect segmentation; outperformed task-specific baselines on unseen data using a curated 30k-image dataset.	
AI Intern (Research & Development) <i>Samsung R&D</i>	Nov. 2021 – Oct. 2022 Bangalore, India
Led R&D for 5 peers on frame-rate conversion; exceeded speed and quality KPIs by 1.5×. 27.9 PSNR achieved on 448x256 video sequences; managed research milestones and senior technical presentations.	

PROJECTS

Media Search Engine <i>Gemini, MongoDB, ChromaDB</i>	Feb. 2026 – Present
- Built a multimodal media retrieval pipeline that indexes image/video assets, extracts metadata, and generates VLM-based descriptions for downstream search. - Designed MongoDB + ChromaDB retrieval architecture with weighted ranking across narrative content, OCR text, and metadata fields.	
LazyResume: OCR & LLM Document Parsing <i>Vision-LLMs, FastAPI</i>	May 2025
- Engineered a pipeline using Vision-based OCR and LLMs; designed to bridge the recruiter-candidate gap by automating unstructured data parsing into structured JSON. - Improved parseability of complex layouts by 20% via custom evaluation hooks for resume-to-JD alignment.	

TECHNICAL SKILLS

Frameworks: Python-PyTorch, TensorFlow, OpenCV, Scikit-Learn, NumPy, Pandas, LangChain, FastAPI
Infrastructure & MLOps: AWS (EC2, DL AMI), Docker, SLURM, Celery, Distributed Training, Quantization
Focus Areas: Deep Learning, Computer Vision, Generative AI, Video Processing, Object Detection